

Thomas Xiao

Irvine, CA | thomasxi@andrew.cmu.edu | LinkedIn | GitHub | Portfolio | Online Resume

EDUCATION

Carnegie Mellon University

B.S. in Mathematical Sciences; Computational and Applied Mathematics

Pittsburgh, PA

Aug. 2026 – May 2030

Arnold O. Beckman High School

High School Diploma; Weighted GPA: 4.63/5.00; SAT: 1570

Irvine, CA

Aug. 2022 – May 2026

EXPERIENCE

Independent Climate Research Collaboration

Research Collaborator

Remote

2024 – 2025

- Analyzed NOAA U.S. climate records from 1895–2024 with Python, harmonic regression, Fourier analysis, wavelets, and change-point detection; built reproducible cleaning, validation, modeling, and visualization workflows.
- Compared frequency-aware models with trend-only baselines across climate variables and regions to identify recurring patterns and structural changes.

USC Marshall School of Business

Research Assistant – Game Design and Consumer Behavior

Remote

2024 – 2025

- Collaborated with Dr. Cosimo Arnesano to analyze BoardGameGeek data with regression and gradient boosting, evaluating ratings, complexity, play time, and consumer response for product-design decisions; presented at CONF-APMM 2025.

Beckman Computer Science Club

Board and Program Lead

Irvine, CA

2024 – 2026

- Taught algorithms and data structures to 15+ students and mentored peers building Python and Discord applications.

PROJECTS

Personal Portfolio and Online Resume | React, JavaScript, HTML/CSS, GitHub Pages

2024 – Present

- Built and deployed a responsive React portfolio by customizing GitProfile with reusable project cards, research pages, theme switching, responsive layouts, and direct links to code, papers, demos, and an online resume.

Desmos Bezier Renderer | Python, Flask, OpenCV, NumPy, Potrace

2023 – Present

- Implemented a Flask image/video processing workflow using Canny edge detection, OpenCV, NumPy, and Potrace to convert frames into Bezier-curve expressions with configurable filtering, export, and browser-based rendering.

M3 Challenge Model | Python, NumPy, pandas, Matplotlib, LaTeX

2025

- Built a reproducible Python model for urban heat, power demand, and community vulnerability, translating assumptions and data into visualizations and a technical paper for the 2025 MathWorks Math Modeling Challenge.

HiMCM Mathematical Model | Python, LaTeX, Mathematical Modeling

2024

- Translated an international real-world modeling prompt into assumptions, quantitative analysis, and a reproducible technical paper; earned HiMCM Honorable Mention.

Mosaic Me | Python, Streamlit, OpenCV, NumPy

2024 – Present

- Extended a Streamlit image-mosaic application with perceptual CIE Lab color matching, image preprocessing, session-state handling, and consolidated download tracking.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, MATLAB, SQL **Frameworks/Tools:** React, Flask, Streamlit, Git, GitHub Pages, LaTeX, Jupyter

Libraries: NumPy, pandas, Matplotlib, Seaborn, OpenCV, scikit-learn, PyWavelets **Methods:** Regression, gradient boosting, Fourier analysis, wavelets, change-point detection, data visualization

Honors: M3 Honorable Mention, 2025 MathWorks Math Modeling Challenge; HiMCM Honorable Mention (top 28% of 1,055 teams); 2nd Place, USC SCMC Math Competition and CSULB Math Day at the Beach